

POKHARA UNIVERSITY

Level: Bachelor

Semester: Fall

Year: 2025

Programme: BCSIT

Full Marks: 100

Course: Computer Architecture and Microprocessor

Pass Marks: 45

Time: 3 hrs.

Candidates are required to answer in their own words as far as practicable. The figures in the margin indicate full marks.

Section "A"

Very Short Answer Questions

Attempt all the questions. [$10 \times 2 = 20$]

1. Define the stored program concept and list its two key characteristics.
2. What is a bidirectional register? Give one example of its use in SAP-2.
3. Define data communication processor and I/O processor.
4. Define the concept of Virtual memory.
5. Calculate the total addressable memory of the 8085 microprocessor, justifying your answer.
6. What is the function of ALE in 8085 microprocessor?
7. How is -5 represented in 1's complement using 4 bits?
8. What do you mean by 16 bit microprocessor?
9. What is the purpose of memory protection in a memory management system?
10. What is a bus system in a computer?

Section "B"

Descriptive Answer Questions

Attempt **any five** questions. [$5 \times 10 = 50$]

11. What is the difference between SAP-1 and SAP-2?
12. What are different types of instructions available in 8085 microprocessor? Write a program to find square of a number stored in C001_H and store the result in C002_H.
13. Perform Multiplication $(-5) \times (-3)$ using booth algorithm.
14. What is virtual memory? Explain address mapping with pages. Also, describe the basic concepts of page fault and page replacement.
15. Explain the following shift micro-operations with examples and diagrams: Logical Shift, Arithmetic Shift, and Circular (Rotate) Shift.

16. Differentiate between a Register Stack and a Memory Stack. Explain the PUSH and POP operations in a memory stack with a suitable diagram.

Group “C”

Long Answer Questions

Attempt **any two** questions. [$2 \times 15 = 30$]

17. a. Explain the DMA operation with Block diagram. [7]
 b. Write down different control signals of 8085 and explain. [8]
18. a. Draw a well-labeled functional block diagram of the Intel 8085 microprocessor and explain the function of each major block. [9]
 b. Draw the block diagram of SAP-1 and list its instruction set along with the corresponding opcodes. [6]
19. a. Describe Flynn's Classification of computer architectures in detail, providing a suitable example for each category. [8]
 b. Draw and explain the timing diagram for the LDA 8000_H instruction. [7]